

# Swathi Kanda

AI Engineer | Machine Learning & Computer Vision | OCR Researcher

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Location: Hyderabad, Telangana, India

## PROFESSIONAL SUMMARY

AI Engineer and Machine Learning professional with an M.Sc. in Statistics and hands-on experience spanning 4 internships and full-time roles in AI development. Specialises in designing end-to-end machine learning pipelines, developing deep learning solutions, and building Computer Vision and OCR systems. Key achievement: engineered OCR pipelines for four low-resource ancient scripts (Brahmi, Sharada, Grantha, Hale Kannada) achieving up to 92% character-level accuracy across 12,000+ training images. Proficient in Python, TensorFlow, OpenCV, SQL, Tableau, and Power BI. Open to roles in AI Engineering, ML Engineering, Data Science, and Computer Vision Research.

## TECHNICAL SKILLS

|                      |                                                                                          |
|----------------------|------------------------------------------------------------------------------------------|
| Machine Learning     | TensorFlow · Scikit-learn · Random Forests · SVMs · Regression · CNNs · RNNs · LSTMs     |
| Computer Vision      | OpenCV · OCR Systems · scikit-image · GIMP · Image Augmentation · Gradio                 |
| NLP & Generative AI  | NLP · Generative AI · Predictive Analytics · Statistical Modelling · Feature Engineering |
| Programming          | Python · SQL (MySQL)                                                                     |
| Data & Visualisation | Power BI · Tableau · Pandas · NumPy                                                      |

## PROFESSIONAL EXPERIENCE

### Artificial Intelligence Engineer

[Vishvam.AI](#) · Serilingampalli, Hyderabad April 2026 – Present

- Architect and deploy Computer Vision and Generative AI solutions within a full-time production engineering capacity.
- Design and ship AI system components, model deployments, and ongoing ML research initiatives.
- Optimise model pipelines for performance and scalability in real-world production environments.

### Software Developer – Artificial Intelligence

[Vishvena Techno Solutions](#) · Hyderabad June 2025 – February 2026

- Engineered end-to-end OCR pipelines for four historical scripts: Brahmi, Sharada, Grantha, and Hale Kannada.
- Generated and processed 12,000+ synthetic and real-world character images using Google Fonts and manuscript PDFs; implemented preprocessing, augmentation, and labelling pipelines.
- Trained TensorFlow-based deep learning models achieving up to ~92% character-level accuracy.
- Segmented manuscript lines and individual characters using GIMP and OpenCV/scikit-image; produced one-to-one script mapping outputs.
- Evaluated models with confusion matrices, precision, recall, and F1-score; deployed interactive OCR demos via Gradio.

### Artificial Intelligence Intern

[Codec Technologies India](#) · Hyderabad March 2025 – April 2025

- Implemented supervised, unsupervised, and reinforcement learning models on real-world datasets.
- Delivered live Perceptron learning sessions with practical demonstrations to peers and stakeholders.
- Built CNN and RNN architectures through lab-based exercises, reinforcing deep learning fundamentals.

### Machine Learning Intern

[Fynn Labs](#) · Hyderabad August 2024 – October 2024

- Conducted market analysis for a fraud detection application, identifying key data patterns and business requirements.
- Developed Random Forest and Logistic Regression models for EV data analysis and predictive insights.
- Collaborated with cross-functional teams to deliver real-world ML solutions from problem scoping to deployment.

## KEY PROJECTS

### Ancient Script OCR System – Brahmi, Sharada, Grantha, Hale Kannada

June 2025 – February 2026 | [Vishvena Techno Solutions](#)  
GitHub: <https://github.com/s199487/vishwena>

Engineered end-to-end OCR pipelines across 4 historical scripts. Generated 12,000+ training images; achieved ~92% character-level accuracy using TensorFlow. Segmented manuscript lines and characters using GIMP and OpenCV/scikit-image; deployed interactive Gradio demo. Technologies: Python, TensorFlow, OpenCV, scikit-image, GIMP, Gradio.

**Vishvena Infotech – AI-Powered Ancient Script OCR & Dataset Generation Platform**

June 2025 – February 2026 | Vishvena Techno Solutions

GitHub: [https://github.com/s199487/vishvena\\_infotech](https://github.com/s199487/vishvena_infotech)

Built automated pipelines for Unicode conversion, synthetic dataset generation, image preprocessing, and multilingual OCR dataset creation. Applied Computer Vision and ML techniques to support historical manuscript digitisation through scalable OCR. Technologies: Python, Computer Vision, OCR, Deep Learning, NLP, Image Processing, Dataset Engineering.

**AI Smart Farming: Precision Agriculture for Sustainable Food Production**

January 2026 – February 2026 | Vishvena Techno Solutions

Live Demo: <https://aismartfarming-156jm2q.gamma.site/>

Developed an AI-driven smart farming platform integrating ML, Computer Vision, and predictive analytics for crop selection, disease identification, yield optimisation, and resource management. Delivered AI-based crop recommendation, early issue monitoring, and data-driven resource insights. Technologies: Python, Machine Learning, Computer Vision, Predictive Modelling, Agricultural Informatics.

**Bharatiya Knowledge Systems (BKS): Digitizing India's Traditional Wisdom Through AI**

January 2026 – February 2026 | Vishvena Techno Solutions

Live Demo: <https://bharatiya-knowledge-syst-rj2705o.gamma.site/>

Designed AI, OCR, and Computer Vision workflows to digitise ancient manuscripts and transform indigenous knowledge into structured, machine-readable formats. Covered domains including agriculture, medicine, mathematics, astronomy, and philosophy. Technologies: Python, AI, Computer Vision, OCR, NLP, Data Analysis, Image Processing, Dataset Engineering.

**Full-Stack Notes Management System**

May 2026 – Present | Vishvam.AI

GitHub (Frontend): <https://github.com/s199487/notes-frontend>

GitHub (Backend): <https://github.com/s199487/notes-backend>

Built a full-stack Notes Management System using Node.js and PostgreSQL. Developed RESTful APIs, designed relational database schemas, and integrated frontend and backend components for a scalable and responsive web application. Technologies: Node.js, PostgreSQL, RESTful APIs.

**Stock Price Prediction & Telecom Churn Prediction**

2024 | Academic Projects

Stock Price Prediction: Built an RNN/LSTM time-series forecasting model on financial data with feature engineering, evaluated using RMSE metrics. Telecom Churn Prediction: Developed a supervised classification model with EDA, class imbalance handling, and full precision/recall/F1 evaluation. Technologies: Python, Keras, Scikit-learn, Pandas.

**E D U C A T I O N**

**Master of Science – Statistics**

Bankatlal Badruka College of Information and Technology, Hyderabad

November 2022 – September 2024

**C E R T I F I C A T I O N S**

AI Tools & ChatGPT Workshop  
MySQL Basics

Data Visualization with Power BI  
Introduction to Data Science

Python Basic Programs